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FACULTY OF ENGINEERING

DEGREE IN ARTIFICIAL INTELLIGENCE ENGINEERING

SOFTWARE ENGINEERING II

INTELLIGENT RECOMMENDATION PLATFORM

USER MANUAL

Version 1.0
Windows and macOS

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A practical guide to installing the platform, finding movie suggestions, recording interests, and fixing common problems.

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1. ABOUT THIS GUIDE

This manual helps you run the project on your own computer and use its movie recommendation page. You do not need to understand programming to follow the main steps. A short technical overview is provided near the end.

The platform suggests movies for an existing numbered user from the included MovieLens sample. It is a local demonstration, not a public streaming service (yet). There is no account creation, movie playback, or personal profile form.

NOTE: The web page itself uses Spanish labels. This English guide keeps those exact labels so you can match each instruction to the screen.

WARNING: The pictures of the web page and consoles in this guide are faithful illustrations based on the project files and verified commands.

2. BEFORE YOU BEGIN

1. Install Python 3.11 or newer. Check the version with py -3 --version in Windows PowerShell, or python3 --version in macOS Terminal. If the command is missing, install Python from python.org.

2. Install Git from git-scm.com and check it with git --version. Git lets you copy the project folder from GitHub.

3. Use a modern web browser. Keep an internet connection for the first package installation. The sample data and prepared model files already come with this repository.

A virtual environment is a small, separate space for this project's Python packages. It prevents these packages from affecting your other Python projects (Python Software Foundation, n.d.).

3. INSTALL AND START ON WINDOWS

3.1 Open PowerShell

Open Windows PowerShell. Move to a folder where you want the project. Then enter the commands below, one line at a time. Git creates a folder named Intelligent-Recommendation-Plataform (Git, n.d.).

```
git clone https://github.com/Michou001/Intelligent-Recommendation-Plataform.git
cd Intelligent-Recommendation-Plataform
py -3 --version
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -e ".[als]"
```

If your Python launcher does not recognize py -3, use python instead, but first confirm it reports version 3.11 or newer. The [als] option installs the default recommendation method; other back-end services are not required for normal local use.

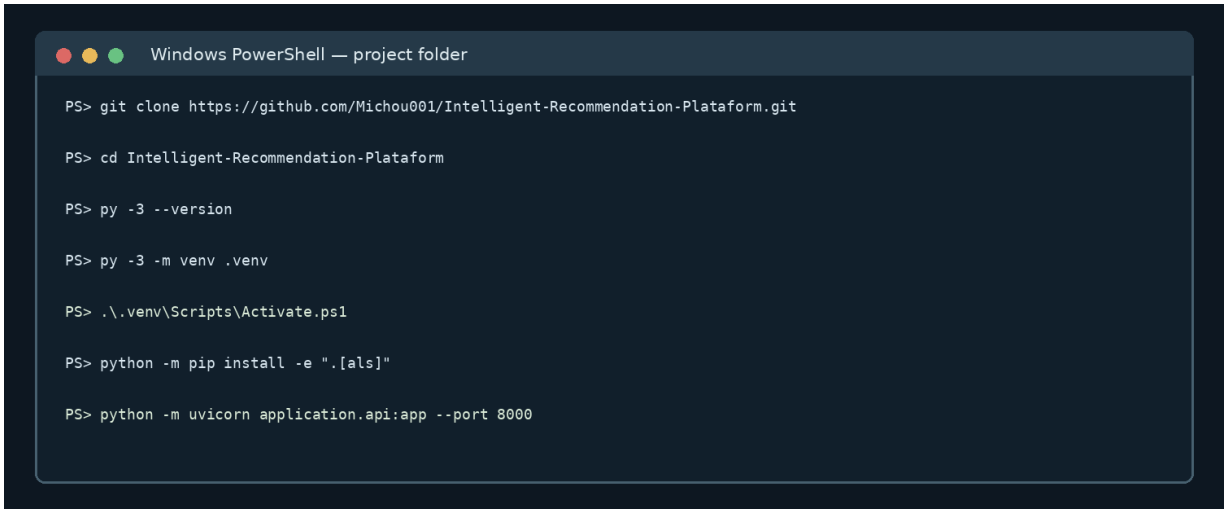


Figure 1. Illustrated PowerShell command sequence; type each line separately.

3.2 Start the service

python -m uvicorn application.api:app --port 8000

Leave this PowerShell window open. In your browser, open http://127.0.0.1:8000/. The address redirects to /ui/. To stop the service, return to PowerShell and press Ctrl+C.

NOTE: If PowerShell blocks Activate.ps1, open Command Prompt in the same project folder and use .venv\Scripts\activate.bat. Then continue with the same pip and uvicorn commands. Do not weaken the system-wide execution policy.

4. INSTALL AND START ON MACOS

Open Terminal and go to a folder where you want the project. Enter the following lines one at a time. A Python version of 3.11 or newer is required.

git clone https://github.com/Michou001/Intelligent-Recommendation-Plataform.git
cd Intelligent-Recommendation-Plataform
python3 --version
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -e ".[als]"
python -m uvicorn application.api:app --port 8000

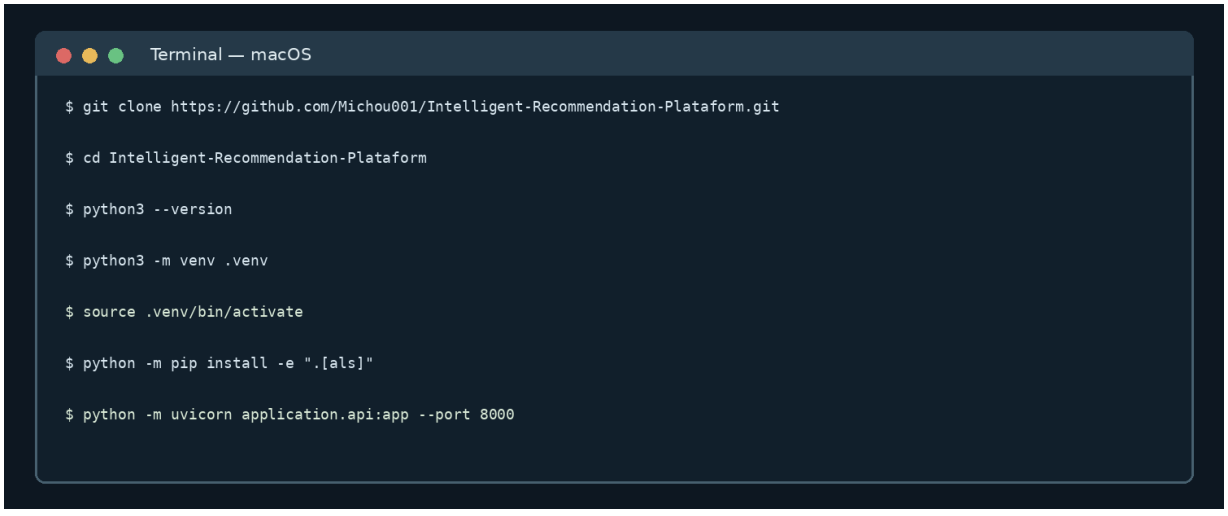


Figure 2. Illustrated macOS Terminal command sequence; type each line separately.

Open http://127.0.0.1:8000/ in a browser. Keep Terminal open while you use the page. Press Control+C in Terminal to stop it.

NOTE: If the optional [als] package cannot be installed on your Mac, use python -m pip install -e . instead. The application may then show a degraded service and use its popularity-based fallback. This still allows a basic demonstration, but it is not the same recommendation method.

5. USE THE WEB PAGE

5.1 Check the service

Open the local address in your browser. At the top right, “servicio operativo” means the service is working. “servicio degradado” means it started but a model or data component needs attention. The screen may show a model name such as “als”.



Figure 3. Illustrated starting screen with the Spanish controls used by the application.

5.2 Ask for recommendations

1. In “Usuario”, type a valid number for an existing sample user. The page begins with 1 when the service responds, the hint below the field shows a user range.

2. In “Cantidad”, choose how many movies to show, from 1 to 100. The starting value is 10.

3. Select “Recomendar”. Read the “Resultados” list below. The movie at rank 1 appears first.

4. If you want a random existing sample user, select “Usuario al azar”. The page chooses a number and runs a new search.

NOTE: These are example users from the sample dataset.

5.3 Understand the results

Each row shows a movie title, possible categories, an order number, a score, and a “Me interesa” button. The score is a relative value used to order this list; it is not a star rating or a guarantee that you will enjoy the movie.

Above the list, “modelo” names the method used. “versión” and “feature store” identify prepared project data; “latencia” is the time spent by the service on this request. These details are informative and do not require action.



Figure 4. Illustrated example with three results for sample user 1; titles come from a local test and values may change.

5.4 Record interest

Select “Me interesa” next to a movie that interests you. If the request is accepted, the button changes to “Registrado”; otherwise it changes to “No registrado”. The event is queued and saved in the background. It does not immediately retrain the model or change the current list (Michou001, 2026).

NOTE: After changing the user number, ask for new recommendations before pressing “Me interesa”. The button records the user number currently shown in the field.

5.5 Example session

For a first test, enter Usuario = 1 and Cantidad = 3, then select Recomendar. Review the three movies and select Me interesa on one of them. Try Usuario al azar for another sample user. Finally, close the browser tab and stop the server with Ctrl+C.

6. BASIC CONFIGURATION AND MAINTENANCE

For ordinary local use, you do not need to edit settings, configure a database server, or run a data-processing command. The repository includes the sample dataset, a prepared feature store, a local database, and model files. By default, it uses local storage and a memory-based event queue.

If you reopen the project later, go to its folder, activate .venv, and run the same uvicorn command again. The site runs only while that command is active. To install a newer project revision, use Git carefully: replacing local files may lose your local changes.

WARNING: Keep the service on your own computer. This guide uses its default local address, 127.0.0.1. Do not expose it to the internet or put private data into the sample project.

6.1 If prepared data is missing

Only use the following recovery commands when the included data or local database files are missing. They rebuild prepared data from the CSV sample and fill the local catalog. Run them inside the project folder with the virtual environment active:

```
python -m data.pipeline
python -m persistence --seed
```

This may take time and may replace generated local data. The normal installation does not need this step. Do not add --download when the included dataset is present.

7. FIX COMMON PROBLEMS

What you see	What to try
Python or Git not found	Install from the official website, reopen the terminal, and check the version again.
PowerShell blocks activation	Use Command Prompt and .venv\Scripts\activate.bat. Leave security policy unchanged.
Page does not open	Keep the server window open. Check for “Uvicorn running on http://127.0.0.1:8000”. If port 8000 is in use, stop the other program or start with --port 8001 and open that port instead.
Service is degraded	Check that artifacts/feature_store/v1 exists. If [als] failed to install, retry that installation or use the basic popularity fallback.
Unknown user or no results	Choose a number shown by the page’s user-range hint. A missing profile is not solved by asking for more movies.
Interest not recorded	Confirm the service is running, select Recomendar again, and then try Me interesa. The button shows Registrado only when the request is accepted.

7.1 Read a warning on screen

If the result includes “Respuesta degradada”, the configured model was not available and the service used the popularity list. You may continue browsing, but the recommendations are less personal. If the page displays “servicio no disponible”, check that the local server is still running.

8. HOW THE SYSTEM WORKS

The browser sends your user number and requested list length to a local service. The service reads prepared movie data, chooses a recommendation method, and returns a ranked list. When you select “Me interesa”, the browser sends a separate interest event for background storage. This is a simple overview rather than a developer guide.

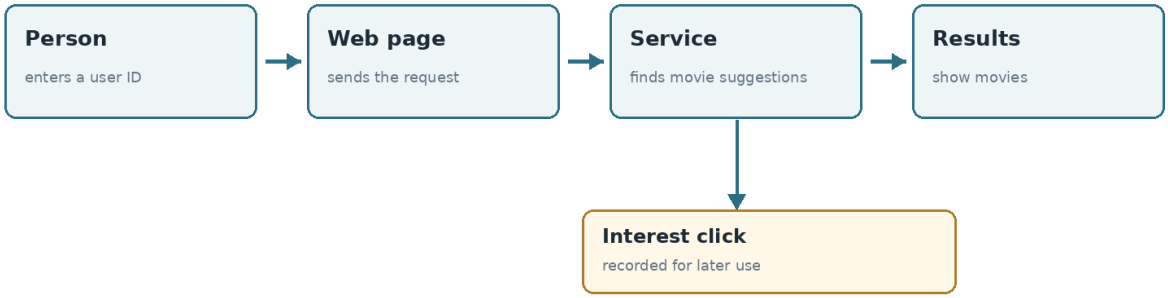


Figure 5. Simplified flow from user action to movie results and interest recording.

The default method is “als”, which uses patterns in earlier movie ratings. If it cannot run, the service can use a popularity list instead. The sample data comes from MovieLens ml-latest-small. The system does not stream films, create new user accounts, or learn from one click immediately.

9. REFERENCES

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